

Qatar-1b

R-2698

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120918 · created 2026-09-01T00:02:42+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456188.66212 +/- 0.00013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.017
Transit depth (Rp/Rs)^2	3.68 +/- 0.66 [%]
Orbital Inclination (inc)	83.38 +/- 1.38
Airmass coefficient 1 (a1)	1.297 +/- 0.02
Airmass coefficient 2 (a2)	-0.21 +/- 0.16
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	None
Optimal Aperture	2.58
Optimal Annulus	9.29
Transit Duration (day)	0.0671 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.017 vs 0.1463 (deviation 2.69 σ)
Duration fit vs published	0.0671 d vs 0.0692 d (deviation 0.24 σ)
Mid-time O-C	-0.3 min
Depth SNR	11.3
Residual scatter	1.51 %

Light curve

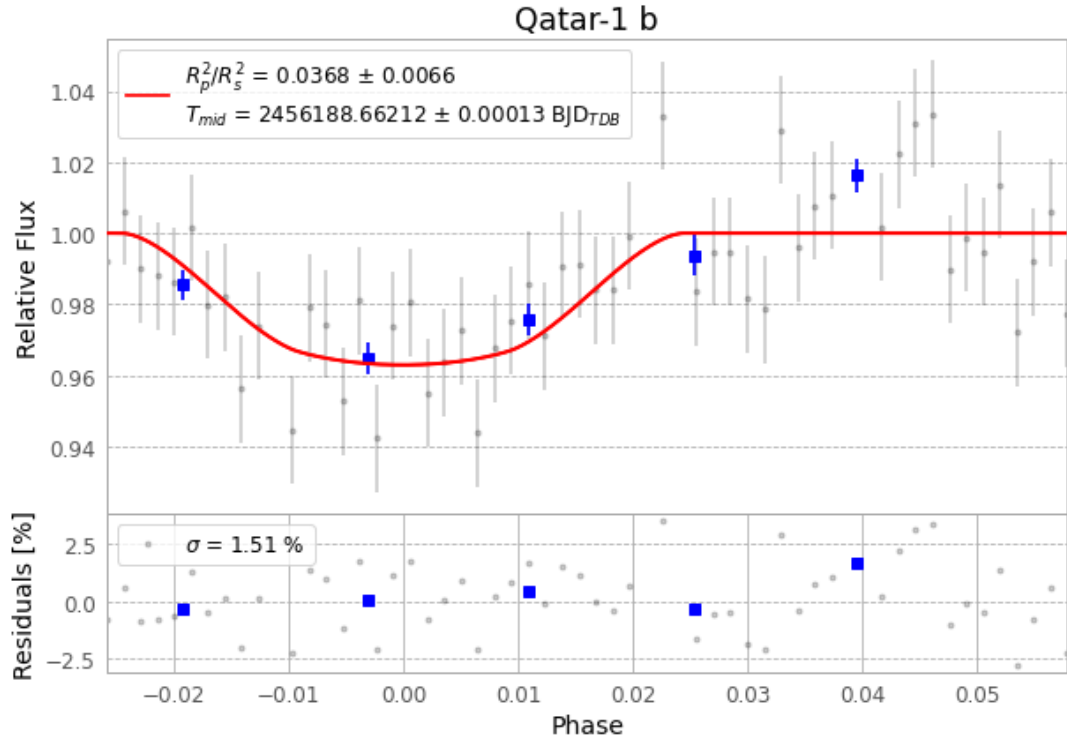


Figure 1 — FinalLightCurve_Qatar-1 b_2012-09-17.png (SHA-256 deb86f42d04acc9c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [413, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 80f41e6fc917efb9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), Qatar-1120918025712.FITS → Qatar-1120918054812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120918.log (d7270e45c54b...), FinalLightCurve_Qatar-1 b_2012-09-17.png (deb86f42d04a...), FinalLightCurve_Qatar-1 b_2012-09-17.csv (97af2d5bed11...)

Compute resources

Wall time 308.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:02Z.